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Title: Return Predictability in Bitcoin ETFs: A Machine Learning

Approach

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Return Predictability in Bitcoin ETFs: A Machine Learning Approach

Abstract

This study employs machine learning classification methods to analyze the predictability of eight prominent crypto exchange-traded funds across four distinct temporal intervals. Numerical experiments employing classification techniques, including Logistic Regression, Linear Regression, Support Vector Machines, Artificial Neural Networks, K-means clustering, and Random Forest algorithms, illustrate the predictability of price fluctuations, both upward and downward. The average classification accuracy of the six algorithms exceeds 50% across all cryptocurrencies and time scales, with Neural Networks and SVM models often surpass 0.80. This indicates a degree of price predictability, suggesting a potential violation of the weak-form Efficient Market Hypothesis (EMH) in crypto ETF markets and the possibility of profitable strategies based on historical data. Among strategies considered, Momentum stands out for those prioritizing returns and willing to tolerate volatility, offering strong growth potential in the crypto space.

Keywords: Crypto ETFs, Return prediction, Machine learning, Efficient market hypothesis, Trading strategies.

1. Introduction

A significant milestone in US cryptocurrency investing was reached on January 10, 2024, when the Securities and Exchange Commission (SEC) approved spot Bitcoin exchange-traded funds (ETFs). These funds allow investors to buy and sell Bitcoin at its current market price. This approval by an important regulator like the SEC is a big step forward. It creates a strong and scalable connection between cryptocurrency and traditional finance, helping to bring the two systems together. Bitcoin ETFs represent a new asset class within the exchange-traded fund framework, offering investors exposure to Bitcoin's potential returns without the complexities of managing

ing cryptocurrency directly. By mirroring traditional investment structures, these ETFs are well-suited for integration into mainstream portfolios, making them accessible to institutional and retail investors. Since their approval in January 2024, Bitcoin ETFs have witnessed a substantial influx of capital, rapidly becoming among the most widely held ETFs.

This study focuses on Spot Bitcoin ETFs, financial instruments that track the price of Bitcoin, a cryptocurrency. These ETFs operate in a way familiar to many investors: Each share of the ETF is backed by actual cryptocurrency held in reserve. This allows investors to follow the price changes of Bitcoin without needing to buy, hold, or manage the cryptocurrency themselves. Spot Bitcoin ETFs are explicitly designed with Bitcoin as the underlying asset and function similarly to traditional ETFs, offering reassurance by operating like conventional ETFs. The availability of eleven Spot Bitcoin ETFs in the US further emphasizes how this new investment option resembles traditional ETFs.

Bitcoin ETFs offer a variety of benefits for both institutional and individual investors. They provide individual investors a straightforward way to gain exposure to Bitcoin, bypassing the complexities of cryptocurrency wallets or exchanges. This simplifies the investment process for many. Additionally, ETFs offer institutions a secure and regulated means to diversify their portfolios with Bitcoin, potentially enhancing their returns. The fact that these ETFs can be traded on traditional stock markets simplifies the process for institutions to incorporate Bitcoin into their existing investment strategies. Notably, the safety and regulation of these ETFs provide reassurance to institutional and retail investors, allowing them to participate confidently in the Bitcoin market.

On the other hand, Spot Bitcoin ETFs are generally considered risky because Bitcoin is inherently volatile, which can lead to significant fluctuations in the ETF's value. This potential for substantial changes should prompt investors to exercise caution and maintain awareness. They must be prepared for significant price fluctuations and the risk of losses. Fees, like expense ratios, are also part of managing ETFs and can affect total returns.

Changes to the regulations could significantly impact the availability of spot Bitcoin ETFs, affecting the investment choices of many investors. Additionally, secure and reliable custody solutions are crucial for managing Bitcoin holdings. Emphasizing the significance of these solutions is critical for fostering investor confidence. Acknowledging that any potential security breaches could harm investor trust is essential. Understanding the tax im-

plications of holding Bitcoin directly, which may vary by region, is crucial for informed decision-making. Investors need to be aware of the unique tax issues in their area.

The existing literature reveals three key gaps. First, there is a lack of research on the efficiency of the market for crypto exchange-traded funds (ETFs). Second, most previous studies relied on standard statistical tests that only assessed whether financial markets are weakly efficient. While these tests can indicate inefficiency, they do not provide insights into capitalizing on market opportunities or quantifying potential profits from consistent trading. Finally, most studies have focused on daily returns in traditional and non-traditional financial markets. However, high-frequency analysis is urgently needed for a more comprehensive understanding of crypto ETFs.

There are important reasons to explore the questions mentioned earlier. Crypto ETFs have strong growth potential and may include more types of digital assets over time. As the market grows, more advanced products—such as leveraged and inverse crypto ETFs—may provide investors with new opportunities to profit from price fluctuations. Some ETFs might also focus on themes like blockchain technology or tokenization. New regulations will play a significant role in shaping these trends and helping to protect investors and stabilize the market.

This study aims to address all previously mentioned issues and answer the question: Can machine learning models accurately predict price movements in crypto ETFs, and how do these models compare across different timeframes? To achieve this, we assess the predictability of returns for eight Bitcoin ETFs by analyzing returns at various intraday frequencies. We utilize techniques such as Support Vector Machines, Logistic Regression, Linear Regression, Artificial Neural Networks, K-means clustering, and Random Forest classification. Our findings indicate that the returns in cryptocurrency ETF markets can be regularly forecasted at daily or minute intervals, achieving classification accuracies reaching up to 89% and an average accuracy of approximately 55% to 60% across all sampled cryptocurrencies. Upon comparing the accuracy metrics among the models, the Neural Networks and SVM models demonstrate the best accuracy, frequently above 0.80, with specific instances attaining 0.86. These models consistently demonstrate strong performance across various time frames and scenarios. The Decision Tree and K-means Clustering models exhibit moderate accuracy, typically ranging from 0.55 to 0.78, rendering them less dependable for high-precision forecasts. Despite challenges with non-linear data patterns, Linear Regression

and Logistic Regression models offer a distinct advantage in interpretability. Their accuracy rates typically range from 0.75 to 0.78.

Our findings, which show a violation of the weak-form efficiency property in crypto ETF markets at daily and various minute levels, have significant implications for investors. We also discuss strategies in crypto ETF markets and conclude that the best approach depends on an investor's risk tolerance and objectives.

The paper is structured as follows: Sections 2 and 3 present the literature review and the data used in this study. Section 4 describes the methodologies used to identify predictability patterns. Sections 5 and 6 provide the test results and examine their robustness. Section 7 discusses various strategies, and finally, Section 8 offers concluding remarks.

2. Literature review

After this brief introduction, it's important to discuss whether the prices of cryptocurrency ETFs can be predicted. Experts are still debating the applicability of the efficient market hypothesis (EMH) to crypto ETFs. However, Investors should exercise caution and take the time to understand the market thoroughly.

There are three types of EMH: weak, semi-strong, and strong. The weak-form states that past prices cannot be used to predict future prices. The semi-strong form states that prices change rapidly due to publicly available information. Moreover, the strong form says prices reflect even private or insider information. The Efficient Market Hypothesis (EMH) asserts that it is impossible to consistently outperform the market average through active management or stock selection, as asset prices rapidly incorporate all available information. According to EMH, any new information is quickly reflected in prices, leaving no room for systematic excess returns. However, the hypothesis has its critics. Some argue that markets are not always perfectly efficient, particularly in the short term, and that skilled investors may take advantage of mispricings or behavioral biases to achieve superior returns.

For example, "Stock Market Prices Do Not Follow Random Walks" by [Lo and MacKinlay \(1988\)](#) shows that stock returns are not random, which contradicts the Random Walks theory. [Poterba and Summers \(1988\)](#) and [Brock et al. \(1992\)](#) demonstrate that the weak-form of the Efficient Market Hypothesis (EMH) is violated in various types of asset returns.

In the context of cryptocurrency markets, [Urquhart \(2016\)](#) finds, in "The Inefficiency of Bitcoin," that Bitcoin prices exhibit serial correlation, further contradicting the weak-form Efficient Market Hypothesis (EMH).

Many studies have subsequently explored the weak-form Efficient Market Hypothesis (EMH) using diverse methodologies, sample frequencies, and benchmark currencies. For example, [Bariviera \(2017\)](#) examines the information efficiency of Bitcoin returns by analyzing their long-range dependence characteristics. [Vidal-Tomas and Ibanez \(2018\)](#) evaluate efficiency in cryptocurrency markets through non-linear statistical approaches. [Sensoy et al. \(2019\)](#) extend the analysis to cross-market cryptocurrency efficiency using multi-fractal analysis. These varied approaches underline the extensive interest in understanding market efficiency in rapidly evolving and diverse financial markets. [Tartakovsky et al. \(2020\)](#) analyze the efficiency of the cryptocurrency market by examining return autocorrelation. Their findings reveal significant short-term inefficiencies, which challenge the Efficient Market Hypothesis (EMH).

[Urquhart and Zhang \(2022\)](#) assess market efficiency using the Adaptive Market Hypothesis (AMH). They demonstrate that the efficiency of Bitcoin and Ethereum fluctuates over time, with active trading strategies occasionally outperforming passive ones.

Research exploring the weak-form Efficient Market Hypothesis (EMH) in ETF markets has revealed significant implications. The findings suggest that inefficiencies exist where historical price data can predict future price movements. [Mazouz and Bowe \(2006\)](#) demonstrate patterns in the price behavior of ETFs that contradict the notion of weak-form efficiency. "Inefficiencies in the Pricing of Exchange-Traded Funds" by [Antti Petajisto \(2017\)](#) examines how ETF prices can deviate from their net asset values (NAVs), particularly in funds holding international or illiquid securities. These collective findings challenge ETF markets' weak-form Efficient Market Hypothesis (EMH), suggesting that past prices may have predictive power, particularly in less liquid or emerging ETF segments.

[Fischer and Krauss \(2021\)](#) investigate the application of machine learning to financial forecasting. They highlight how these methods can improve predictive accuracy and expand the capabilities of traditional econometric models. Their findings suggest that machine learning techniques, such as deep learning, significantly enhance price forecasts, providing a strong foundation for exchange-traded fund (ETF) forecasting.

[Shih et al. \(2024\)](#) compare deep learning methods to the Fama-French

three-factor model for predicting ETF returns. Their research supports the effectiveness of deep learning in ETF price forecasting, revealing that Neural Networks—especially Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN)—outperform traditional models by effectively identifying complex market patterns.

Due to their recent emergence, examining weak-form market efficiency in Bitcoin ETFs is still in early development. However, insights from broader cryptocurrency markets suggest that these financial instruments may exhibit inefficiencies. Factors such as limited market depth, evolving regulatory frameworks, and high volatility can contribute to deviations from the Random Walks hypothesis. This, in turn, suggests a potential for predictability in price movements, offering a more optimistic outlook for investors and enabling the possibility of more informed trading decisions.

3. Data

Spot and Futures Crypto ETFs are two distinct vehicles that provide exposure to cryptocurrencies. A Spot Crypto ETF directly invests in the underlying cryptocurrency, such as Bitcoin or Ethereum, and holds the actual asset securely in custody. This means the ETF's price closely tracks the real-time market value of the cryptocurrency it represents. Spot ETFs are popular among long-term investors seeking a direct way to gain exposure to crypto without the complexity of managing wallets or private keys. However, they face significant regulatory hurdles in many regions, which investors should be aware of, and risks like custody breaches are inherent due to the need to store physical assets.

On the other hand, Futures Crypto ETFs do not invest in physical cryptocurrencies; instead, they trade in futures contracts tied to them. These contracts are agreements to buy or sell the asset at a specified price on a future date. Futures ETFs have a more transparent and straightforward launch process, making them easier to implement. They often follow a more explicit regulatory path, exemplified by Bitcoin Futures ETFs that have already received approval in specific markets. However, pricing discrepancies can occur because futures contracts do not always match spot market prices. This misalignment is mainly due to market conditions such as contango or backwardation. Additionally, regularly rolling over contracts to maintain continuous exposure can introduce transaction costs and potential slippage, which may erode returns over time.

The primary difference between the two lies in their underlying assets and investment strategies. Spot ETFs offer direct exposure by holding the underlying cryptocurrency, making them ideal for investors seeking straightforward, long-term exposure to crypto markets. Futures ETFs, in contrast, cater to more sophisticated traders looking to leverage or hedge their positions without holding the actual asset. The increasing popularity of spot crypto ETFs makes this study particularly relevant. Spot ETFs provide simplicity and direct exposure to the underlying assets, making them a valuable tool for understanding the dynamics of these increasingly important financial products.

This study utilizes dollar-denominated crypto ETF data obtained from the Bloomberg terminal. Our dataset covers January 11, 2024, to August 16, 2024. During this period, we selected eight widely traded cryptocurrencies denominated in US dollars, which significantly impacted the market. Table 1 lists the ticker symbols and fund names.

Table 1: The list of Spot Crypto ETFs from January 11 to August 16, 2024.

Ticker	Fund name
ARKB	ARK 21Shares Bitcoin ETF
BITB	Bitwise Bitcoin ETF Trust
BRRR	Coinshares Valkyrie Bitcoin Fund
BTCO	Invesco Galaxy Bitcoin ETF
EZBC	Franklin Bitcoin ETF
FBTC	Fidelity Wise Origin Bitcoin Fund
HODL	VanEck Bitcoin ETF
IBIT	iShares Bitcoin Trust ETF

Figure 1 shows the daily closing prices for each of the selected crypto ETFs. The performance of various crypto ETFs highlights the diverse characteristics, risks, and rewards associated with these instruments. One notable aspect of crypto ETFs is their high volatility. The chart reveals significant price fluctuations across all ETFs, reflecting the inherent instability of the cryptocurrency market. Some ETFs, such as "HODL," exhibit more pronounced peaks and dips, suggesting aggressive or leveraged exposure to crypto assets. This volatility highlights the potential for substantial gains

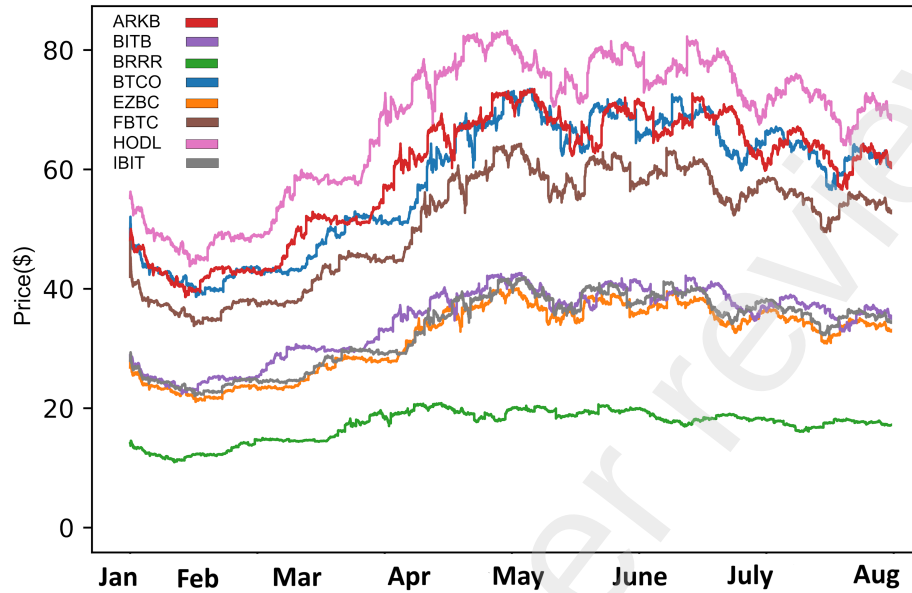


Figure 1: Closing daily prices of eight crypto ETFs from January 11 to August 16, 2024

and significant losses, serving as a reminder of the market's unpredictability and the importance of exercising caution in investment decisions.

Another observation is the diverse performance of the ETFs. While all appear influenced by the broader crypto market, the degree of growth and fluctuation varies. For example, "HODL" shows higher peaks and outperformance during market rallies, whereas "BRRR" maintains a relatively flat trajectory, indicating a more conservative or niche strategy. Additionally, the correlation among the ETFs is evident, as they generally follow similar trends, rising and falling in tandem with the broader cryptocurrency market. This alignment suggests that macro-level market events, such as Bitcoin price surges or regulatory developments, significantly impact all crypto ETFs, influencing their performance to varying degrees.

Tables 2, 3, 4, and 5 provide descriptive statistics for the log returns across multiple timeframes, including 15 minutes, 30 minutes, 60 minutes, and daily. With the highest mean and median values across all timeframes, HODL consistently outshines in terms of average performance, indicating its strong relative performance. However, BRRR consistently has the lowest mean and median, suggesting a less successful or conservative strategy. The middle ground is occupied by ARKB, BTCO, and FBTC, which offer a performance

that falls between the extremes of HODL and BRRR.

According to the standard deviation, HODL is the most volatile ETF, with notable price fluctuations throughout all periods. This high volatility suggests a higher-risk, higher-reward profile. On the other hand, because it is conservative and reliable, BRRR is the least volatile. ETFs with moderate volatility, such as ARKB and BTCO, appeal to investors seeking a strategy that balances risk and reward.

HODL has the highest maximum value when analyzing price ranges, peaking at 83.04 in the daily data. This underlines its strong potential during bullish market conditions. In contrast, BRRR has the lowest minimum value, reinforcing its conservative approach. FBTC and BTCO maintain relatively high maximum prices compared to their minimums, indicating meaningful upward price movements while avoiding extreme lows.

Examining upward and downward movements, HODL has the highest proportion of upward movements, reaching 55% in specific time frames, which highlights its bullish momentum. However, its higher rate of downward movements in shorter time frames suggests susceptibility to corrections. Most other ETFs exhibit balanced movement percentages, hovering around 50%, reflecting their alignment with broader market trends. However, BRRR displays a slightly bearish outlook with a lower upward movement percentage of 49% in shorter periods.

Shorter intervals (15-60 minutes) show greater price fluctuations, underscoring the intraday volatility typical of crypto markets. The statistics stabilize over longer time frames (daily), providing a clearer picture of the ETFs' long-term performance. HODL continues to dominate in mean values, maximum prices, and upward momentum, making it an attractive option for long-term, risk-tolerant investors. Meanwhile, BRRR maintains its stability, offering minimal price swings but limited growth potential.

	Mean	Median	Std	Min	Max	Up Move	Up Move %	Down Move	Down Move %
ARKB	60.61	63.36	8.89	39.3	72.97	77.0	50.0	72.0	47.0
BITB	33.16	34.6	4.77	21.4	39.7	74.0	49.0	76.0	50.0
BRRR	17.51	18.04	2.16	11.27	20.66	80.0	52.0	69.0	45.0
BTCO	61.04	63.47	8.83	39.29	73.07	77.0	50.0	73.0	48.0
EZBC	35.47	36.78	4.83	23.31	42.35	77.0	50.0	73.0	48.0
FBTC	53.23	55.43	7.74	33.87	63.74	76.0	50.0	74.0	49.0
HODL	69.45	71.96	9.55	44.8	82.86	84.0	55.0	66.0	43.0
IBIT	34.6	36.03	5.04	22.39	41.88	77.0	50.0	73.0	48.0

Table 2: Summary statistics of 15-minute log-returns from January 11 to August 16, 2024. 'Up Move' and 'Down Move' indicate the number of positive and negative returns, respectively.

	Mean	Median	Std	Min	Max	Up Move	Up Move %	Down Move	Down Move %
ARKB	60.69	63.33	8.96	39.0	73.1	457.0	50.0	451.0	49.0
BITB	33.19	34.59	4.82	21.13	40.02	462.0	49.0	458.0	49.0
BRRR	17.53	18.04	2.17	11.1	20.73	244.0	47.0	260.0	50.0
BTCO	61.01	63.5	8.83	39.01	73.27	252.0	50.0	246.0	49.0
EZBC	35.48	36.85	4.87	22.64	42.45	211.0	50.0	205.0	48.0
FBTC	53.18	55.43	7.74	33.99	64.14	498.0	50.0	468.0	47.0
HODL	69.51	71.97	9.51	44.38	82.97	245.0	48.0	250.0	49.0
IBIT	34.64	36.09	5.04	22.18	41.81	486.0	49.0	480.0	49.0

Table 3: Summary statistics of 30-minute log-returns from January 11 to August 16, 2024. 'Up Move' and 'Down Move' indicate the number of positive and negative returns, respectively.

	Mean	Median	Std	Min	Max	Up Move	Up Move %	Down Move	Down Move %
ARKB	60.68	63.36	8.97	38.65	73.37	922.0	50.0	885.0	48.0
BITB	33.19	34.6	4.82	21.13	40.02	916.0	49.0	902.0	48.0
BRRR	17.53	18.03	2.17	11.04	20.79	521.0	50.0	477.0	46.0
BTCO	61.02	63.6	8.83	39.01	73.27	512.0	51.0	482.0	48.0
EZBC	35.5	36.85	4.87	22.64	42.45	421.0	50.0	408.0	48.0
FBTC	53.19	55.44	7.74	33.97	64.2	998.0	51.0	940.0	48.0
HODL	69.51	71.98	9.51	44.28	82.99	507.0	50.0	487.0	48.0
IBIT	34.65	36.1	5.05	22.16	41.84	987.0	50.0	935.0	48.0

Table 4: Summary statistics of 60-minute log-returns from January 11 to August 16, 2024. 'Up Move' and 'Down Move' indicate the number of positive and negative returns, respectively.

	Mean	Median	Std	Min	Max	Up Move	Up Move %	Down Move	Down Move %
ARKB	60.68	63.36	8.97	38.65	73.37	1842.0	50.0	1758.0	48.0
BITB	33.19	34.6	4.82	21.13	40.02	1859.0	50.0	1782.0	48.0
BRRR	17.53	18.02	2.16	11.04	20.79	1004.0	49.0	959.0	46.0
BTCO	61.03	63.59	8.83	38.72	73.27	1012.0	50.0	976.0	48.0
EZBC	35.51	36.85	4.87	22.43	42.55	843.0	50.0	809.0	48.0
FBTC	53.19	55.46	7.74	33.84	64.2	1963.0	50.0	1884.0	48.0
HODL	69.51	72.02	9.53	43.9	83.04	1001.0	50.0	981.0	49.0
IBIT	34.64	36.12	5.05	22.07	41.84	1934.0	49.0	1895.0	48.0

Table 5: Summary statistics of daily log-returns from January 11 to August 16, 2024. 'Up Move' and 'Down Move' indicate the number of positive and negative returns, respectively.

3.1. Measures

We use thirty-four features (Table 6) to predict the daily and minute-level returns in the next period. The fundamental features are the open, close, high, low prices, and high-low range. The second group of features includes the last five lagged log returns. The exponentially weighted moving average of the close prices, the cumulative sum of the log returns for the previous 3 and 5 days, and their differences are used as ten additional features. Two relative strength indexes, two rate of change indexes, and the weighted moving average of close prices are also utilized as features. The relative strength index is calculated using the following formula:

$$RSI = 100 - \frac{100}{1 + \frac{SMMA(U,n)}{SMMA(D,n)}} \quad (1)$$

The smoothed moving average SMMA is an exponential moving average of the upward and downward price changes in the last n trading days. The upward and downward price changes are defined as follows:

$$D = \begin{cases} \text{Close}_{\text{previous}} - \text{Close}_{\text{now}}, & \text{if } \text{Close}_{\text{previous}} > \text{Close}_{\text{now}} \\ U = 0 & \end{cases} \quad (2)$$

$$U = \begin{cases} \text{Close}_{\text{now}} - \text{Close}_{\text{previous}}, & \text{if } \text{Close}_{\text{now}} > \text{Close}_{\text{previous}} \\ D = 0 & \end{cases} \quad (3)$$

After identifying upward and downward price changes, the exponential moving average is computed over the predefined last n trading days. Our feature set uses two distinct n values: 9 and 14 days. Once the index's relative strength has been determined, four additional indicator elements are developed to indicate whether the assets are overbought or oversold for the 9— and 14-day RSI values. RSI(14) is the default setting recommended by [J. Welles Wilder \(1986\)](#), the developer of the Relative Strength Index. On the other hand, RSI(9) is commonly favored by traders seeking shorter-term signals. These values are considered “industry standard,” making them a natural starting point for analysis by practitioners and researchers.

The Rate of Change (RoC) is valuable for understanding market momentum and price dynamics. It helps traders assess market strength, identify potential trend reversals, and determine optimal entry and exit points. The Rate of Change is a momentum indicator that measures the percentage change in an asset's price over a specific period. This indicator enables

traders and analysts to assess the strength of price movements and identify overbought or oversold conditions. The formula is as follows:

$$\text{ROC} = \frac{P_t - P_{t-n}}{P_{t-n}} \times 100 \quad (4)$$

The Exponential Weighted Moving Average (EWMA) is a type of moving average that assigns greater weight to recent data points while gradually decreasing the influence of older data. Unlike the simple moving average, which gives equal weight to all observations within the window, the EWMA uses a smoothing factor (λ) to prioritize recent prices or values. The formula for EWMA is:

$$\text{EWMA}_t = \lambda P_t + (1 - \lambda) \text{EWMA}_{t-1} \quad (5)$$

The Momentum Indicator is a simple yet effective tool used in technical analysis to measure the rate of price change over a specified period. It helps traders identify the speed of price movement and assess whether the trend is gaining or losing strength, which can indicate potential reversals or continuations.

$$\%R = \frac{\text{High}(n) - \text{Last Close}}{\text{High}(n) - \text{Low}(n)} \quad (6)$$

The Aroon Stochastic Oscillator is a hybrid technical indicator that combines the Aroon Indicator and the Stochastic Oscillator to assess trend strength and direction while detecting overbought and oversold conditions. The Aroon indicator assesses the duration since the highest high and lowest low within a specified time frame, indicating whether an asset is trending or oscillating within a range. The Stochastic Oscillator evaluates the closing price's position relative to its recent high-low range, providing insights into momentum and potential reversal points. The Aroon Stochastic Oscillator thoroughly assesses trend strength and momentum, allowing traders to detect trend reversals and choose optimal entry or exit points.

The Double Exponential Moving Average (DEMA) is a refined moving average designed to minimize the lag associated with traditional moving averages, making it more responsive to price changes. Unlike the Simple Moving Average (SMA) or Exponential Moving Average (EMA), the DEMAs combines two EMAs to achieve a smoother and faster-reacting indicator. The formula is:

$$\text{DEMA}_t = 2\text{EMA}_t - \text{EMA}(\text{EMA}_t) \quad (7)$$

A comprehensive elucidation and compilation of formulas for many technical indicators are available in [Kyoung-jae et al. \(2000\)](#).

4. Empirical analysis

This section computes target returns using four distinct time intervals: daily, 15-minute, 30-minute, and 60-minute frequencies. Based on the calculated returns at various time horizons, a binary classification problem is formulated. Specifically, the binary target variable is assigned a value of 1 if the return at the subsequent time step is positive, and -1 if the return is negative. Six classification algorithms are implemented to classify these target variables: Logistic Regression, Linear Regression, Support Vector Machines (SVM), Artificial Neural Networks (ANN), K-means clustering, and Random Forests. These models are developed using Python's widely adopted scikit-learn library.

We briefly overview the classification algorithms and their results without going into technical implementation details. In addition to machine learning models, we incorporate traditional forecasting methods, such as the Random Walk model and ARIMA, as benchmarks. Finally, we use the t-test and the Wilcoxon signed-rank test to compare the performance of the different methods statistically.

4.1. Machine learning methods

Classification is a fundamental task in machine learning to predict a discrete class label for input data. It is a part of supervised learning, where models are trained using labeled datasets. The objective is to identify patterns or decision boundaries in the data, enabling the model to assign new, unseen data points to the correct categories. Essentially, classification involves mapping input features to predefined class labels.

Accuracy is one of the most commonly used metrics for evaluating the performance of a classification model. It measures the proportion of correctly classified instances from the total number of cases in the dataset.

This study examined six distinct classification algorithms to categorize target variables at different temporal frequencies. We utilized Python's popular scikit-learn package to implement various classification techniques, including logistic regression, linear regression, Support Vector Machines, Artificial Neural Networks, Random Forests, and the K-means algorithm. While

Table 6: Set of features utilized in the classification algorithms

Feature name	Number of lags/window size	Number of features
Open, high, low, close	Last one period	4
High-low	Last one period	1
Returns ($r_{t-1}, \dots, r_{t-5}$)	Last five periods	5
Moving average (MA)	Last 5 days	1
Correlation MA and close	Last one value	1
$\Sigma_k = \sum_{t=k}^N r_{t-1} + \dots + r_{t-k},$ $k = 3, 5$	Last 3/5 days	2
$\Sigma_5 - \Sigma_3$	Last 3/5 days	1
Relative strength index (RSI)	Window size = 6, 14	2
$\mathbf{1}(RSI_6) < 20\%, \mathbf{1}(RSI_{14}) < 20\%$	Buy signals w.r.t. the RSIs	2
$-\mathbf{1}(RSI_6) > 80\%, -\mathbf{1}(RSI_{14}) > 80\%$	Sell signals w.r.t. the RSIs	2
Moving average convergence divergence (MACD)	Fast period = 5, slow = 10, signal period = 5	3
Rate of change, rate of change return	Window = 9, 14	2
Exponential weighted moving average (EWMA)	$\lambda = 0.9$	1
Momentum indicator	Window = 5	1
Average true range	Window = 5, 10	2
Williams' %R	Window = 14	1
Aroon stochastic oscillator	Window = 14	1
Double exponential moving average (DEMA)	Window = 10 15	1

existing research provides a detailed discussion of these algorithms, this section will focus on their application.

4.1.1. Logistic regression

Logistic Regression is a foundational algorithm in machine learning designed primarily for classification tasks. Despite its name, it is not a regression algorithm for predicting continuous values but a method for predicting the probability of a discrete outcome. It is particularly effective in binary classification problems, where the goal is to categorize data into one of two possible classes.

Logistic Regression is a supervised learning algorithm widely used for binary classification tasks. It estimates the probability that a given input belongs to a specific class by applying the logistic (sigmoid) function to a linear combination of input features. Based on this probability, the model categorizes the input into one of two classes.

Python provides several libraries for implementing the popular machine learning algorithm, Logistic Regression. Among these, scikit-learn is one of the most widely used, and we will focus on it.

The results for the Logistic Regression are given in Table 7. The "Daily-0.8" shows the best overall performance for most crypto ETFs. According to the table, the "Daily-0.8" configuration yields the highest overall performance for most ETFs. For example, ARKB achieves a performance value of 0.74, BRRR reaches 0.85, and EZBC achieves 0.81, indicating strong predictive capability at this temporal resolution and threshold. This suggests that daily aggregated data provide more consistent and reliable inputs for the Logistic Regression model, likely because they capture broader trends and smooth out noise that may be present in more granular time intervals.

As the time interval decreases to 60 minutes, 30 minutes, and 15 minutes, the performance generally declines, indicating that shorter intervals may introduce more variability in the data. For instance, ARKB's performance drops from 0.74 in "Daily-0.8" to 0.58 in "60min-0.8" and further to 0.55 in "30min-0.9." Similar trends are observed for other ETFs, such as BITB and HODL, suggesting that the increased frequency of observations at shorter intervals may reduce the Logistic Regression model's ability to discern meaningful patterns.

Overall, ETFs like BRRR and BTCO consistently exhibit high-performance metrics, indicating their predictive strength and reliability in Logistic Regression. In contrast, ETFs like FBTC and IBIT exhibit lower performance

metrics across most configurations, with values of 0.52 for FBTC and 0.56 for IBIT in the "Daily-0.8" scenario.

	ARKB	BITB	BRRR	BTCT	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.74	0.56	0.85	0.85	0.81	0.52	0.78	0.56
60 min-0.9	0.6	0.64	0.78	0.69	0.78	0.59	0.71	0.54
60 min-0.8	0.58	0.65	0.86	0.74	0.78	0.56	0.74	0.46
30 min-0.9	0.55	0.59	0.82	0.68	0.84	0.51	0.72	0.49
30 min-0.8	0.58	0.61	0.84	0.74	0.78	0.53	0.81	0.52
15 min-0.9	0.55	0.59	0.78	0.74	0.77	0.52	0.74	0.52
15 min-0.8	0.61	0.61	0.82	0.75	0.82	0.51	0.8	0.56

Table 7: Logistic regression prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

4.1.2. Support vector machine

Support Vector Machines (SVMs) are powerful and versatile machine-learning algorithms for classification and regression tasks. They are particularly effective for high-dimensional data and are widely employed in various applications. The primary objective of SVMs is to find the optimal decision boundary, also known as the hyperplane, that best separates data points belonging to different classes. At its core, an SVM attempts to find a hyperplane in an N -dimensional space, where N is the number of features, that separates the data into distinct classes. The optimal hyperplane maximizes the margin, the distance between it and the nearest data points from each class. These closest points, known as support vectors, support the definition of the hyperplane.

Mathematically, the optimization problem in SVM aims to maximize the margin while ensuring the correct classification of the data points. This results in a decision boundary that generalizes well to unseen data. Given the training vectors x_i for $i = 1, 2, \dots, N$ with a sample size of N observations, the support vector machine classification algorithm solves the following problem given by

$$\min_{w, \eta} \quad \frac{1}{2} w^T w + C \sum_{i=1}^N \eta_i \quad (8)$$

$$\text{subject to} \quad y_i(w^T \phi(x_i)) \geq 1 - \eta_i, \quad i = 1, 2, \dots, N \quad (9)$$

$$\eta_i \geq 0, \quad i = 1, 2, \dots, N \quad (10)$$

The optimization problem in Equation 8 can be globally solved using the Karush–Kuhn–Tucker (KKT) conditions. Detailed derivations are provided in [Huang et al. \(2007\)](#).

The results in Table 8 indicate that the "Daily-0.8" yields the highest overall performance across most ETFs. For instance, ARKB achieves 0.70, BRRR reaches 0.85, and EZBC achieves 0.81, highlighting that daily aggregated data provides more stable and reliable patterns for the SVM model to classify effectively. This performance may be attributed to the reduced noise in daily data compared to shorter time intervals, allowing the SVM to focus on more prominent and consistent trends in the cryptocurrency data.

As the time interval decreases from daily to 60-minute, 30-minute, and 15-minute resolutions, there is a noticeable decline in performance across most ETFs. For example, ARKB drops from 0.70 in "Daily-0.8" to 0.59 in "60min-0.8" and further to 0.55 in "15min-0.9." This trend suggests that shorter intervals may introduce variability or noise, challenging the SVM's ability to identify meaningful patterns. However, some ETFs, such as BRRR and BTCO, maintain relatively high performance even at shorter intervals, demonstrating their improved predictive ability and less susceptibility to the data's enhanced sensitivity.

	ARKB	BITB	BRRR	BTCT	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.7	0.52	0.85	0.85	0.81	0.59	0.81	0.48
60 min-0.9	0.59	0.64	0.78	0.69	0.78	0.58	0.71	0.55
60 min-0.8	0.59	0.65	0.86	0.74	0.78	0.55	0.74	0.48
30 min-0.9	0.56	0.59	0.82	0.68	0.84	0.48	0.72	0.5
30 min-0.8	0.58	0.61	0.84	0.74	0.78	0.53	0.81	0.54
15 min-0.9	0.55	0.59	0.78	0.74	0.77	0.53	0.74	0.53
15 min-0.8	0.61	0.61	0.82	0.75	0.82	0.51	0.8	0.56

Table 8: Support vector machine (SVM) prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

4.1.3. Random Forest classification

Random Forest is a robust and flexible machine-learning algorithm for classification and regression tasks. It belongs to the family of ensemble learning methods, which combines multiple Decision Trees to achieve better performance than a single tree. In the case of classification, Random Forest creates an ensemble of Decision Trees during training and aggregates their outputs to make the final prediction. This majority voting mechanism helps to improve accuracy and robustness while reducing the risk of over-fitting, a common issue with individual Decision Trees.

The efficacy of Random Forest is attributed to its incorporation of randomness through two primary mechanisms: bootstrap sampling and random feature selection. Bootstrap sampling, also known as bagging, involves training each tree on a randomly selected subset of the training data, obtained by sampling with replacement. This ensures that each tree in the forest is exposed to different data points, leading to diversity among the trees. Additionally, at each split in a tree, Random Forest considers only a random subset of features rather than all available features. This further enhances diversity and reduces the correlation between trees, making the ensemble more effective in capturing patterns in the data. More details can be found at [Breiman, L. \(2001\)](#).

To implement Random Forest in Python, libraries such as scikit-learn provide an intuitive interface for training and evaluating the model. The

process typically involves splitting the data into training and testing sets, fitting the Random Forest classifier to the training data, and assessing its performance on the test data. Metrics such as accuracy, precision, recall, and F1 Score are commonly used to evaluate the quality of a model. Additionally, feature importance ratings can be obtained to identify the variables most significantly influencing the predictions.

Table 9 presents the results of a Random Forest classification model applied to various cryptocurrency ETFs, such as ARKB, BITB, BRRR, BTCO, EZBC, FBTC, HODL, and IBIT, over different periods (Daily, 60 minutes, 30 minutes, and 15 minutes) and threshold levels (0.8 and 0.9). For the "Daily-0.8", the Random Forest model exhibits its strongest performance for several ETFs, including BRRR (0.70), BTCO (0.63), and EZBC (0.74). These results suggest that daily aggregated data provides more stable and predictable trends for the Random Forest algorithm, likely due to the reduced noise and greater signal strength. Conversely, the performance for ETFs like BITB and IBIT under "Daily-0.8" is relatively low, at 0.44 and 0.41, respectively. This disparity indicates that certain ETFs may require additional feature engineering or alternative modeling approaches to improve prediction accuracy.

Moving from daily to 60-minute, 30-minute, and 15-minute intervals, the performance generally declines for most exchange-traded funds (ETFs). For example, ARKB drops from 0.63 in "Daily-0.8" to 0.55 in "60min-0.8" and further to 0.51 in "15min-0.9." A similar trend is observed for other ETFs, such as BTCO and EZBC, which experience declines across shorter intervals. This drop in performance suggests that higher-frequency data may introduce more noise, which can obscure meaningful patterns that Random Forest relies on for classification. However, BRRR shows a slightly better performance at the 60-minute interval with thresholds of 0.8 and 0.9, achieving 0.78 and 0.74, respectively, highlighting its robustness to temporal changes compared to other ETFs.

	ARKB	BITB	BRRR	BTCO	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.63	0.44	0.7	0.63	0.74	0.56	0.63	0.41
60 min-0.9	0.57	0.54	0.74	0.49	0.62	0.49	0.65	0.47
60 min-0.8	0.55	0.6	0.78	0.6	0.66	0.56	0.65	0.46
30 min-0.9	0.55	0.47	0.68	0.58	0.66	0.49	0.62	0.51
30 min-0.8	0.52	0.52	0.76	0.63	0.64	0.51	0.73	0.54
15 min-0.9	0.51	0.57	0.71	0.6	0.64	0.51	0.58	0.46
15 min-0.8	0.51	0.52	0.68	0.62	0.7	0.51	0.66	0.53

Table 9: Random Forest prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

4.1.4. Using Linear regression for classification

Linear Regression is often used to forecast continuous variables, such as the price or return of a financial asset. It can also be used to predict price direction by modeling the correlation between independent variables (features) and the dependent variable, representing the asset's future price or return. The regression output is a continuous variable denoting the anticipated price or return.

Linear Regression can forecast price movement direction (up or down), with continuous predictions subsequently categorized into binary classifications. If the expected amount exceeds 0, it may signify an upward trend (price increase). If the projected number is zero or less, it may indicate a downward trend, characterized by a price decline. This transformation converts the issue into a binary classification task, emphasizing the direction of price movement rather than the precise anticipated value. After this, metrics such as accuracy and precision can be used to evaluate the model's ability to predict the direction of prices. Table 10 displays the performance of Linear Regression predictions converted into a binary classification of price direction (up or down) across various cryptocurrency ETFs.

In the "Daily-0.8" model, the best performance is observed for several ETFs, such as BRRR (0.85) and BTCO (0.85), indicating that the Linear Regression predictions effectively capture daily price movements for these assets. For BRRR and BTCO, the model correctly predicted price direction 85% of the time at the daily level of data. This suggests that daily

aggregated data enable the model to capture the underlying trends in price direction more effectively. However, other ETFs, such as BITB (0.37) and IBIT (0.59), exhibit significantly lower performance under the same conditions, suggesting that the model struggles to accurately predict the direction of price movements for these assets.

The overall performance of most ETFs tends to decline in shorter time intervals, such as 60-minute, 30-minute, and 15-minute data. For example, ARKB achieves an accuracy of 0.67 under "Daily-0.8," which drops to 0.56 and 0.51 under "30min-0.9" and "15min-0.9," respectively. This decline suggests that the Linear Regression model becomes less reliable at predicting price direction as the time interval shortens, likely due to increased noise and variability in high-frequency data. Similarly, while BRRR performs well under "60min-0.8" (0.86), its performance also decreases at shorter intervals, reflecting the challenges of predicting direction in noisier, shorter-term data.

	ARKB	BITB	BRRR	BTCO	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.67	0.37	0.85	0.85	0.81	0.52	0.78	0.59
60 min-0.9	0.54	0.63	0.78	0.69	0.78	0.58	0.71	0.51
60 min-0.8	0.59	0.65	0.86	0.74	0.78	0.59	0.74	0.42
30 min-0.9	0.56	0.59	0.82	0.68	0.84	0.49	0.72	0.49
30 min-0.8	0.56	0.6	0.84	0.74	0.78	0.54	0.81	0.48
15 min-0.9	0.56	0.59	0.78	0.74	0.77	0.52	0.74	0.53
15 min-0.8	0.61	0.61	0.82	0.75	0.82	0.51	0.8	0.53

Table 10: Linear Regression prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

4.1.5. Artificial Neural Networks

The multilayer perceptron is a widely utilized and adaptable design of Neural Networks. A multilayer perceptron can approximate diverse functions (see [Principe et al. \(1999\)](#)). The multilayer perceptron's capacity to capture nonlinearity is facilitated by employing smooth activation functions that interlink several layers, with prevalent options being the logistic and hyperbolic tangent functions. Moreover, each element of a specific layer

influences all components of the subsequent layer. Multilayer Perceptrons (MLPs) are typically trained using the back-propagation algorithm. The back-propagation algorithm transmits faults throughout the network, facilitating the adjustment of the hidden processing elements. The MLP is trained using an error-correcting backpropagation algorithm that requires knowledge of the system's desired response. Kara et al. (2011) provides an example of the successful application of Artificial Neural Networks in the stock market.

This study employs the multilayer perceptron (MLP) model, a type of Artificial Neural Networks. In the model, we use two hidden layers containing thirty-four and two nodes, respectively, denoted as (34, 2). The number of hidden layers is sufficient to identify potential non-linear relationships among the input features, while the number of nodes aligns with the number of features examined.

The values in Table 11 reflect the accuracy of the Neural Networks in predicting the price direction (e.g., "up" or "down") for each ETF. At the "Daily-0.8" setting, the Neural Networks achieves its best performance for ETFs such as BRRR (0.85), BTCO (0.85), and EZBC (0.81), suggesting that daily aggregated data provides the model with clear trends and patterns to learn from, resulting in accurate predictions for these assets. This high performance demonstrates the Neural Networks' capability to generalize well when the noise is reduced. In contrast, the model's performance is relatively weak for ARKB (0.44) and IBIT (0.44), suggesting that these ETFs may exhibit more volatile or irregular price patterns, which makes them more challenging for the model to predict accurately.

In the shorter intervals, such as 60-minute, 30-minute, and 15-minute data, the performance metrics generally decline for most ETFs, reflecting the challenges of dealing with noisier and more variable data. For instance, ARKB's performance improves slightly from 0.44 under "Daily-0.8" to 0.59 under "60min-0.8" but then drops to 0.43 under "15min-0.8." Similarly, while BRRR achieves strong results under "Daily-0.8" and "60min-0.8" (0.85 and 0.86, respectively), its performance decreases to 0.78 and 0.82 under "15min-0.9" and "15min-0.8," respectively. This trend highlights the Neural Networks' challenges in maintaining high accuracy as data granularity increases, likely due to the emergence of increased noise and fewer predictable patterns over short intervals.

	ARKB	BITB	BRRR	BTCO	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.44	0.56	0.85	0.85	0.81	0.7	0.81	0.44
60 min-0.9	0.58	0.64	0.78	0.69	0.78	0.59	0.71	0.45
60 min-0.8	0.59	0.65	0.86	0.74	0.78	0.55	0.74	0.49
30 min-0.9	0.56	0.59	0.82	0.68	0.84	0.48	0.72	0.49
30 min-0.8	0.58	0.61	0.84	0.74	0.78	0.51	0.81	0.53
15 min-0.9	0.55	0.59	0.78	0.74	0.77	0.53	0.74	0.54
15 min-0.8	0.43	0.61	0.82	0.75	0.82	0.51	0.8	0.57

Table 11: Artificial Neural Networks prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

4.1.6. *K-means classification*

K-means classification is a semi-supervised approach that modifies the traditional K-means clustering algorithm for classification tasks. While K-means typically divides data into K clusters based on their similarity, it can be adapted to assign class labels by associating each cluster with a predefined category.

In this method, the dataset is first grouped into clusters so that similar data points are grouped together. Each point is assigned to the cluster whose centroid is closest, often determined using a distance metric such as Euclidean distance. After clustering is completed, class labels are assigned to each cluster through a majority-voting process, based on the known labels of the training samples within that cluster. See [Bock, H. \(2008\)](#) for further details.

An illustrative implementation of K-means classification in Python involves partitioning a dataset into K clusters and assigning labels to the clusters based on a majority vote. Upon training, the model can categorize additional data points based on their proximity to the selected cluster centroids. The precision of the classification depends on the alignment of the clusters with the actual class boundaries, which the structure and separability of the data can influence. K-means classification is not a replacement for supervised learning models, such as Logistic Regression or Neural Networks; however, it is a valuable method in situations with insufficient labeled data or where exploratory data analysis is necessary. Its simplicity, efficiency,

and versatility render it an excellent instrument for diverse semi-supervised learning tasks.

Table 12 represents the results of applying the K-means clustering algorithm to various cryptocurrency ETFs. The K-means method demonstrates commendable performance for specific ETFs in the "Daily-0.8" analysis, including ARKB (0.70), BRRR (0.81), and BTCO (0.74), suggesting that daily data reveals distinct patterns for effectively clustering these assets into significant categories. This indicates that the daily aggregation effectively captures the dominant trends in the data, enabling the K-means algorithm to form clusters that closely correspond to the actual class labels. However, the performance for ETFs like BITB (0.48) and IBIT (0.44) is significantly lower, which may indicate noisier data or weaker feature relationships that make clustering less effective for these assets.

The performance of most ETFs often declines when the period is decreased to shorter intervals, such as 60-minute, 30-minute, and 15-minute data. For instance, ARKB's accuracy declines from 0.70 in "Daily-0.8" to 0.55 in "60min-0.8" and subsequently to 0.53 in "15min-0.8." Similar trends are observed for BRRR, where the performance drops from 0.81 in "Daily-0.8" to 0.55 in "60min-0.8" and 0.51 in "15min-0.8." This decline underscores the growing challenge of clustering high-frequency data, where noise and short-term volatility can obscure the underlying relationships between features. Such challenges emphasize K-means' limitations in handling highly granular data without further preprocessing or feature engineering.

A closer examination of individual ETFs reveals significant variability in their performance. BRRR and BTCO consistently achieve higher clustering accuracy, suggesting that these ETFs exhibit more distinct and predictable patterns that K-means can effectively capture. In contrast, BITB and IBIT consistently perform poorly in almost all periods, indicating a need for more effective clustering methods due to their data's lack of clear separability. This variability underscores the importance of understanding the unique characteristics of each ETF and adjusting the clustering process accordingly, such as by optimizing the number of clusters (K) or incorporating domain-specific features.

4.2. Time series methods

In finance, Random Walk and ARIMA are popular techniques for predicting prices and simulating time series data. A Random Walk assumes that future price changes are entirely unknown and occur according to a

	ARKB	BITB	BRRR	BTCO	EZBC	FBTC	HODL	IBIT
Daily-0.8	0.7	0.48	0.81	0.74	0.74	0.56	0.52	0.44
60 min-0.9	0.44	0.43	0.56	0.55	0.48	0.54	0.49	0.55
60 min-0.8	0.55	0.55	0.55	0.44	0.46	0.51	0.56	0.52
30 min-0.9	0.51	0.51	0.48	0.53	0.51	0.5	0.6	0.46
30 min-0.8	0.48	0.48	0.51	0.54	0.52	0.49	0.73	0.45
15 min-0.9	0.52	0.48	0.38	0.5	0.51	0.49	0.52	0.49
15 min-0.8	0.53	0.49	0.51	0.54	0.52	0.49	0.53	0.53

Table 12: K-means prediction accuracy across various time intervals and thresholds from January 11 to August 16, 2024.

stochastic process in which the changes are equally distributed and independent of each other. According to the Efficient Market Hypothesis, historical pricing data cannot be used to predict future prices. However, the ARIMA (Auto-Regressive Integrated Moving Average) approach considers characteristics such as autocorrelation and trends to make forecasts. ARIMA models are well-suited for financial time series that exhibit stationarity or can be made stationary through differencing. Table ref tab:12 shows the performance of the Random Walk and ARIMA methods for forecasting different cryptocurrency ETFs (ARKB, BITB, BRRR, BTCO, EZBC, FBTC, HODL, and IBIT) across various time scales.

For the Random Walk method, performance remains relatively consistent across different periods, with values clustering around 0.62 for shorter intervals, such as 15-minute and 30-minute scales, at a threshold of 0.8. Notably, the method performs best at the "Daily-0.8" for several ETFs, including ARKB, BRRR, and IBIT, achieving a peak performance of 0.62. This consistency aligns with the assumption that price movements follow a stochastic process and are difficult to predict beyond their randomness. The uniformity in performance across different cryptocurrencies suggests that the Random Walk method is largely unaffected by the specific characteristics of each asset, as it assumes no underlying trends or patterns.

On the other hand, the ARIMA method exhibits more significant performance variability, indicating its dependence on accurately locating trends and autocorrelations in the time series data. For some ETFs, such as BRRR,

ARIMA performs reasonably well, especially at shorter time intervals, like "15 min-0.8," where it achieves a performance of 0.62. Other ETFs, such as ARKB and FBTC, perform worse, with values ranging from 0.47 to 0.50 over various periods. This suggests that ARIMA struggles to identify significant patterns for specific cryptocurrencies. This could be due to high volatility or the lack of stationarity in the data, a crucial prerequisite for ARIMA models.

In conclusion, the Random Walk approach performs better across a range of ETFs and is more reliable and consistent in its erratic nature. However, due to its sensitivity to the statistical characteristics of the data, ARIMA is less dependable overall, even though it exhibits promise for specific assets and periods. These findings suggest that ARIMA can be beneficial when trends or autocorrelations are present, particularly in short-term forecasting scenarios. In contrast, a Random Walk is better suited for general forecasting in volatile markets.

	ARKB	BITB	BRRR	BTCO	EZBC	FBTC	HODL	IBIT
Random Walk								
15 min-0.8	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62
15 min-0.9	0.56	0.56	0.56	0.56	0.57	0.56	0.57	0.56
30 min-0.8	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61
30 min-0.9	0.54	0.55	0.55	0.58	0.55	0.55	0.54	0.54
60 min-0.8	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61
60 min-0.9	0.54	0.55	0.55	0.58	0.55	0.55	0.54	0.54
Daily-0.8	0.62	0.61	0.62	0.62	0.61	0.61	0.62	0.62
Daily-0.9	0.57	0.58	0.57	0.57	0.58	0.58	0.57	0.57
ARIMA								
15 min-0.8	0.47	0.62	0.47	0.48	0.47	0.62	0.48	0.48
15 min-0.9	0.53	0.53	0.53	0.53	0.56	0.53	0.57	0.56
30 min-0.8	0.5	0.5	0.5	0.49	0.5	0.5	0.5	0.61
30 min-0.9	0.56	0.56	0.56	0.57	0.55	0.55	0.56	0.54
60 min-0.8	0.5	0.5	0.61	0.61	0.61	0.5	0.61	0.61
60 min-0.9	0.56	0.56	0.56	0.54	0.56	0.56	0.56	0.54
Daily-0.8	0.52	0.51	0.52	0.52	0.51	0.5	0.52	0.51
Daily-0.9	0.54	0.44	0.54	0.54	0.44	0.44	0.54	0.53

Table 13: Performance of Random Walk and ARIMA time series forecasting over different crypto ETFs and different time scales from January 11 to August 16, 2024.

5. Comparative analysis

This section compares the implemented methods using ROC curves and different tests. The ROC curves compare the performance of several models based on their ability to classify outcomes accurately. According to Figure 2, the Neural Networks model achieved the highest AUC (0.60), indicating it performed better than the others, though its performance is still moderate. Logistic Regression and SVM both had AUCs around 0.53, showing only a slight improvement over random guessing. Linear Regression and Decision Tree models also performed similarly with AUCs around 0.52. KMeans, ARIMA, and RandomWalk models also performed similarly with AUCs around 0.52.

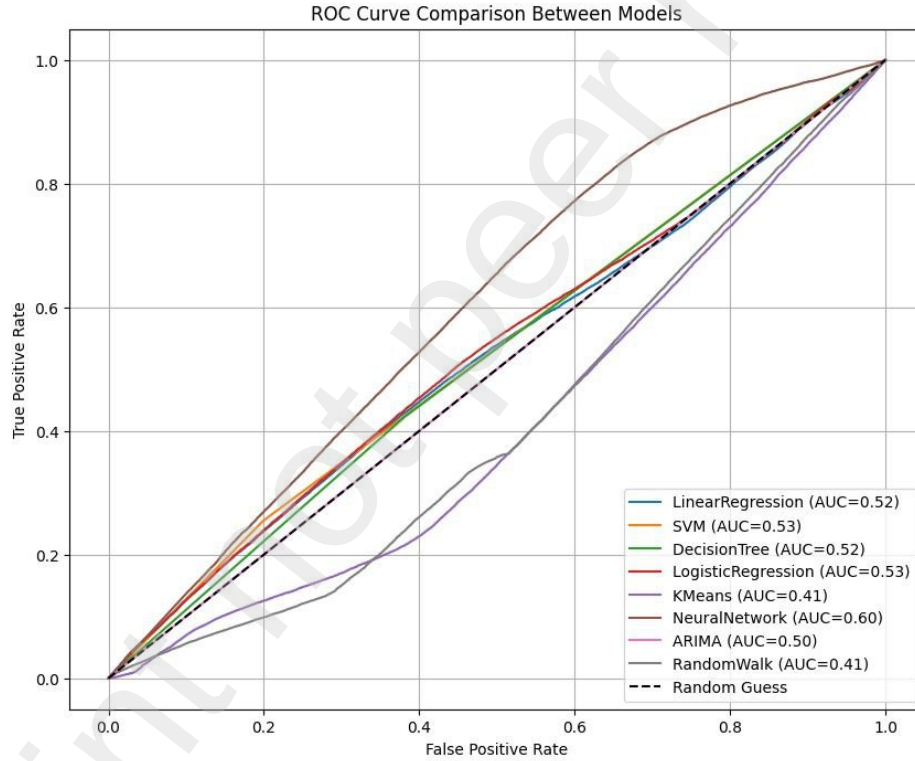


Figure 2: Comparison of model performance using ROC curves from January 11 to August 16, 2024.

Tables A.1, A.2, and A.3 in the Appendix provide a comparative analysis using t-tests to evaluate ARIMA against several machine learning models and benchmarks. This analysis covers various time scales, including 15 minutes, 30 minutes, 60 minutes, 6 hours, and daily intervals.

The results in Table A.1 show that ARIMA consistently outperforms the Random Walk benchmark, as well as Neural Networks, K-means, Logistic Regression, Decision Trees, SVM, and Linear Regression. The t-values, which are predominantly negative across all time scales, highlight the statistical significance of ARIMA's superior performance. With significance levels often at 1%, ARIMA, for example, shows a decisive advantage over Neural Networks and K-means, further validating its statistical significance. The reliability of ARIMA is further supported by its superiority over Logistic Regression and Decision Trees, being evident across time intervals, with most results being significant at the 1% or 5% levels.

Table A.2 reveals a consistent pattern where SVM generally outperforms K-means across all time scales, with positive t-values that are significant at the 1% level, reflecting a robust statistical difference. However, when SVM is compared to Logistic Regression, Decision Trees, and other models, the t-values are close to zero and often not statistically significant, indicating little to no difference in performance between these methods in most cases.

Comparisons with K-means demonstrate significantly higher performance for Linear Regression, with positive t-values consistently significant at the 1% level. This indicates a clear advantage of Linear Regression over K-means. In contrast, the comparisons between Linear Regression and Logistic Regression, Neural Networks, and Decision Trees yield t-values near zero, with no significant differences observed across the time scales. This suggests that Linear Regression and these models yield similar results for the given tasks.

As shown by the positive t-values significant at the 1% level, the Table A.3 findings consistently favor Logistic Regression when compared to K-means across all time scales. This suggests that Logistic Regression outperforms K-means by a significant and reliable margin. With t-values that are positive and statistically significant, Decision Trees also perform noticeably better than K-means at most time scales, demonstrating their superiority over K-means.

Nevertheless, when comparing Neural Networks and K-means, the t-values are negative and not statistically significant. This indicates no difference in the performance of the two models across the tested scenarios, suggesting that K-means and Neural Networks perform similarly on various classification tasks.

The t-values for Decision Trees and Neural Networks are comparable, and no statistical significance is seen over time scales. This suggests that the

performance of these two models is similar. Similarly, as the t-values stay close to zero and are not statistically significant throughout all periods, there are no discernible performance differences between Logistic Regression and Neural Networks.

The Tables A.4, A.5 and A.6 present the results of the Wilcoxon signed-rank test, which compares the performance of ARIMA and several classification algorithms such as Logistic Regression, Decision Tree, SVM, Linear Regression, Neural Networks over different time scales, including daily, 15 minutes, and 30 minutes. The p-values and signed-rank statistics are provided for each comparison, with significance levels denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

In Table A.4, ARIMA consistently outperforms Random Walk across all time scales, as indicated by statistically significant results at most significance levels. This highlights ARIMA's ability to capture patterns that extend beyond baseline random processes. Compared to Logistic Regression, Linear Regression, and SVM, ARIMA's superiority is reflected in significant p-values for many time intervals, suggesting its robust predictive performance relative to these algorithms. However, the comparison with Decision Trees and Neural Networks reveals mixed results, with some time scales showing significant differences while others do not, indicating context-dependent performance.

In Table A.5, SVM consistently produces tiny p-values, indicating that it performs much better than K-means on all time scales. Likewise, SVM performs strongly across temporal resolutions, outperforming Logistic Regression in numerous cases. Less consistency exists in the results when comparing Decision Trees and Linear Regression; some time scales show statistically significant differences, while others do not, indicating context-dependent performance advantages. Results from comparing Neural Networks with Linear Regression are mixed, with notable differences at specific time scales. This suggests that, under some circumstances, Neural Networks may be particularly effective at modeling temporal dependencies. These findings highlight the differences in relative performance among various models, which vary depending on the time scale and data properties. Specifically, SVM typically demonstrates superior performance compared to alternative approaches.

The results in Table A.1 show that ARIMA consistently outperforms the Random Walk benchmark, as well as Neural Networks, K-means, Logistic Regression, Decision Trees, SVM, and Linear Regression. The t-values, which are predominantly negative across all time scales, highlight the statistical significance of ARIMA's superior performance. With significance levels often at

1%, ARIMA, for example, shows a decisive advantage over Neural Networks and K-means, further validating its statistical significance. The reliability of ARIMA is further supported by its superiority over Logistic Regression and Decision Trees, being visible over time intervals, with most results being significant at the 1% or 5% levels.

Table A.2 reveals a consistent pattern where SVM generally outperforms K-means across all time scales, with positive t-values that are significant at the 1% level, reflecting a robust statistical difference. However, when SVM is compared to Logistic Regression, Decision Trees, and other models, the t-values are close to zero and often not statistically significant, indicating little to no difference in performance between these methods in most cases.

The comparison against K-means shows significantly higher performance for Linear Regression, with positive t-values consistently significant at the 1% level. This indicates a clear advantage of Linear Regression over K-means. In contrast, the comparisons between Linear Regression and Logistic Regression, Neural Networks, and Decision Trees yield t-values near zero, with no significant differences observed across the time scales. This suggests that Linear Regression and these models yield similar results for the given tasks.

As shown by positive t-values significant at the 1% level, the Table A.3 findings consistently favor Logistic Regression when compared to K-means across all time scales. This suggests that Logistic Regression outperforms K-means by a significant and reliable margin. With t-values that are positive and statistically significant, Decision Trees also perform noticeably better than K-means at most time scales, demonstrating their superiority over K-means.

Nevertheless, when compared, the t-values for Neural Networks and K-means are not statistically significant. This indicates no difference in the performance of the two models across the tested scenarios, suggesting that K-means and Neural Networks perform similarly on various classification tasks.

The t-values for Decision Trees and Neural Networks are nearly equal, and no statistical significance is seen over time scales. This suggests that the performance of these two models is similar. Similarly, as the t-values stay close to zero and are not statistically significant throughout all periods, there are no discernible performance differences between Logistic Regression and Neural Networks.

Tables A.4, A.5 and A.6 present the results of the Wilcoxon signed-rank

test, which compares the performance of ARIMA and several classification algorithms such as Logistic Regression, Decision Tree, SVM, Linear Regression, Neural Networks over different time scales, including daily, 15 minutes, and 30 minutes. The p-values and signed-rank statistics are provided for each comparison, with significance levels denoted by *, **, and *** at the 10%, 5%, and 1% levels, respectively.

In Table A.4, ARIMA consistently outperforms Random Walk across all time scales, as indicated by statistically significant results at most significance levels. This highlights ARIMA's ability to capture patterns that extend beyond baseline random processes. Compared to Logistic Regression, Linear Regression, and SVM, ARIMA's superiority is reflected in significant p-values for many time intervals, suggesting its robust predictive performance relative to these algorithms. However, the comparison with Decision Trees and Neural Networks reveals mixed outcomes, with some time scales showing significant differences while others do not, indicating context-dependent performance.

In Table A.5, SVM consistently produces tiny p-values, indicating that it performs much better than K-means on all time scales. Likewise, SVM performs strongly across temporal resolutions, outperforming Logistic Regression in numerous cases. Less consistency exists in the results when comparing Decision Trees and Linear Regression; some time scales show statistically significant differences, while others do not, indicating context-dependent performance advantages. Results from comparing Neural Networks with Linear Regression are mixed, with notable differences at specific time scales. This suggests that, under certain circumstances, Neural Networks may be particularly well-suited for capturing intricate temporal patterns. These findings highlight the differences in relative performance among various models, which vary depending on the time scale and data properties. Specifically, SVM typically demonstrates superior performance compared to alternative approaches.

Statistically significant differences identified by low p-values and high significance levels in Table A.6 demonstrate that Neural Networks regularly perform better than the alternative approaches in almost all cases. This suggests that Neural Networks outperform other dataset configurations regarding performance and generalization.

Though it performs marginally worse than Neural Networks, Logistic Regression comes second, routinely exceeding K-means and consistently outperforming Decision Trees. Neural Networks have a distinct advantage over

Logistic Regression, as evidenced by the frequently statistically significant differences between the two.

Although there are notable differences in many comparisons, K-means outperforms Decision Trees in most cases; however, it is still less efficient than Neural Networks and Logistic Regression. Decision Trees continually perform worse than all other methods, making them the poorest approach in this examination. These results demonstrate that Neural Networks are the most reliable and successful classification technique in this assessment, followed by Decision Trees, K-means, and Logistic Regression.

6. Robustness analysis

Random smoothing is a popular probabilistic machine learning strategy for enhancing classifier robustness, particularly against adversarial attacks. Adding random noise to the input and aggregating predictions across several noisy samples transforms any base classifier into a certifiably robust one. The central concept is building a "smoothed" classifier that generates the most likely prediction by applying the underlying classifier's majority vote to the noisy inputs. This technique guarantees that small input changes, including those caused by adversarial attacks, are unlikely to change the model's final forecast.

First, we subject the input data to random noise, typically Gaussian or uniform. The predictions made by the base classifier are then assessed by the smoothed classifier using several noisy iterations of the same input. The smoothed classifier produces a robust final result by obtaining the majority vote across these predictions. Additionally, randomized smoothing provides a formal robustness certification of resilience by guaranteeing that the classifier's prediction remains stable within a perturbation radius of the input. This radius depends on several variables, including the model's prediction confidence and the kind and magnitude of the noise distribution.

Randomized smoothing offers several essential applications. It is instrumental in defending against adversarial attacks, where minor alterations to input data can deceive conventional machine-learning models. By demonstrating resilience, randomized smoothing ensures that a model's predictions stay consistent despite such attacks. Additionally, this method aids in assessing prediction uncertainty by analyzing the variability introduced by noise.

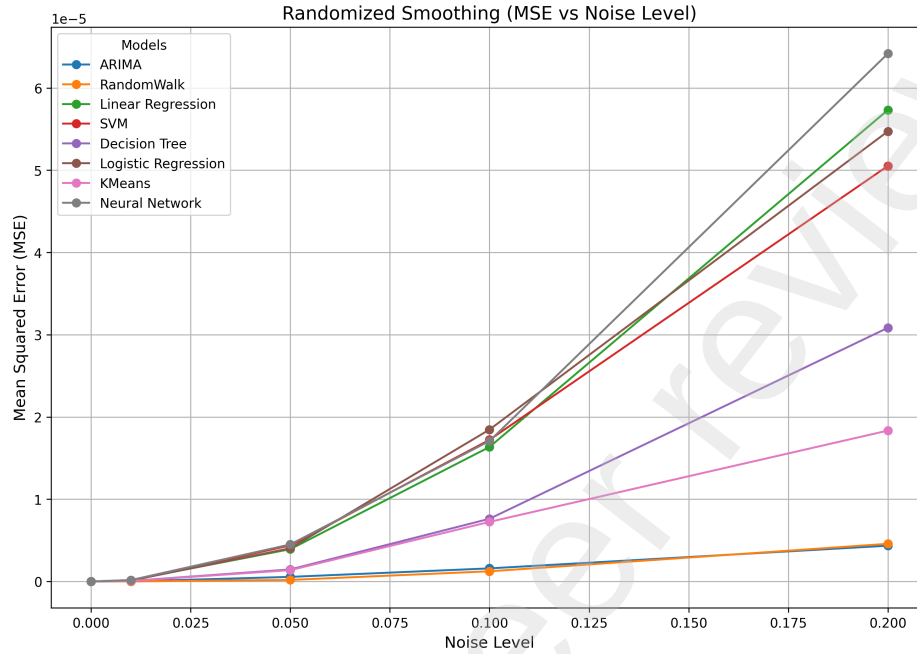


Figure 3: Randomized smoothing analysis from January 11 to August 16, 2024.

The theoretical foundation for randomized smoothing was formalized in research such as "Certified Defenses for Adversarial Examples via Randomized Smoothing" by [Cohen et al. \(2019\)](#), which established its effectiveness as a defense mechanism. Since then, further work has explored variations of noise distributions and applications beyond adversarial robustness.

Figure 3 shows the relationship between the noise level and the mean squared error (MSE) for various machine learning models under the randomized smoothing framework. The key aspects of the results can be interpreted as follows:

As the noise level increases, all models' mean squared error (MSE) generally rises, indicating that higher levels degrade prediction accuracy. Among the models, K-means and Logistic Regression exhibit significantly lower MSE across the range of noise levels, suggesting these methods are more robust to noise. ARIMA and Random Walk, being traditional time-series models, maintain relatively stable MSEs under lower noise levels but diverge at higher noise levels.

Conversely, Neural Networks and SVMs experience the steepest growth

in MSE as noise levels increase, highlighting their sensitivity to perturbations introduced by randomized smoothing. This aligns with their known susceptibility to adversarial perturbations, given their lack of robust defense mechanisms.

The standout performance of Logistic Regression and K-means underlines their suitability for robust tasks, especially when randomized smoothing is applied. Conversely, the degradation in Neural Networks' performance highlights the need for further robustness enhancements in deep learning models for noisy environments.

7. Trading strategies

Financial markets enable investors to buy and sell various assets, profiting from price fluctuations. These fluctuations can be much more significant in the digital currency market than in other financial markets. Despite employing various methods, such as technical analysis, accurately identifying cryptocurrency trends remains a challenging task.

However, these fluctuations are significantly more pronounced in cryptocurrency markets than in other financial markets. Despite various methods, such as technical analysis, it is not easy to accurately identify the trend of cryptocurrencies. The price of a cryptocurrency either increases or decreases. Therefore, profiting might seem easy in theory. However, the most critical point that many investors miss is determining the optimal price to make a purchase and the optimal price to sell the purchased cryptocurrency. One way to mitigate such risks is to employ effective trading strategies.

The buy-and-hold strategy is a popular and passive approach to investing in cryptocurrencies. In this strategy, investors buy cryptocurrencies and hold them for an extended period, regardless of short-term market fluctuations. This approach relies more on fundamental analysis, emphasizing cryptocurrencies' long-term value and growth potential, rather than focusing on technical indicators or daily fluctuations. The primary objective of the hold strategy is to capitalize on the long-term appreciation of cryptocurrencies. In other words, investors can reduce risk by avoiding hasty decisions, such as selling low-priced assets or buying at peak prices.

Dollar-cost averaging (DCA) is another strategy that enables you to invest in the market regularly and gradually, focusing on long-term benefits. It involves dividing your investment into smaller portions and injecting them into the market regularly, such as weekly or monthly. This method has

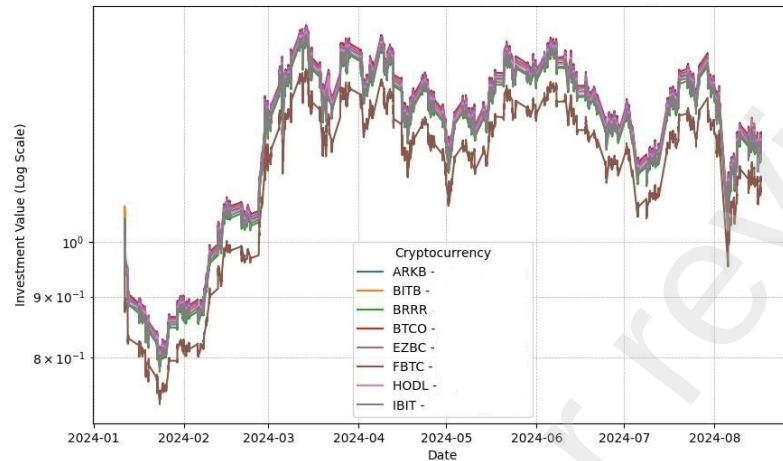


Figure 4: Comparison of Buy and Hold strategy of eight crypto ETFs from January 11 to August 16, 2024.

several significant advantages. First, it allows you to buy at regular intervals, regardless of whether prices are high or low, thereby reducing the impact of short-term fluctuations on your average cost. Second, it eliminates the need for daily analysis or detailed forecasts, allowing you to invest automatically with a more optimistic outlook.

The last one is Momentum trading in cryptocurrency markets. This strategy is based on the principle that assets performing well will continue to do so. In contrast, those underperforming or showing weaker performance compared to the market average will likely continue declining. This approach relies on identifying trends and capitalizing on market movements driven by investor sentiment, technical patterns, and trading volume. At its core, momentum trading involves recognizing assets that exhibit strong upward or downward trends. Traders, armed with technical indicators such as moving averages, the Relative Strength Index (RSI), and the Moving Average Convergence/Divergence (MACD), can gauge the trend's strength and identify potential entry or exit points, thereby gaining valuable insights. Refer to 'Evaluating Trading Strategies' by [Harvey et al.\(2014\)](#).

Figures 4, 5, and 6 illustrate the Buy-and-Hold, DCA, and Momentum strategies for cryptocurrency ETFs. The Momentum strategy emerges as the most profitable, dynamically shifting investments toward the best-performing

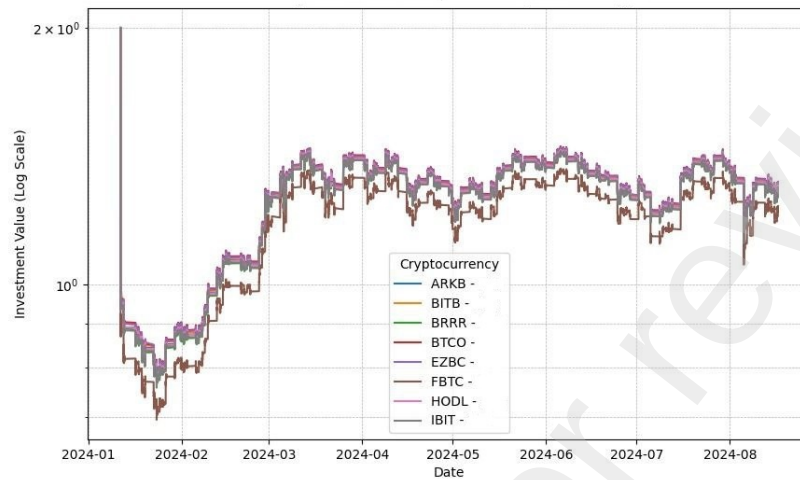


Figure 5: Comparison of DCA strategy of eight crypto ETFs from January 11 to August 16, 2024.

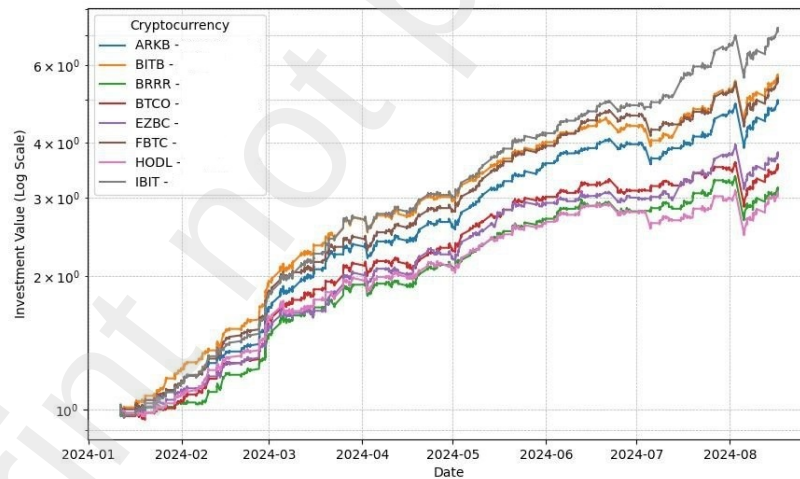


Figure 6: Comparison of the Momentum strategy of eight crypto ETFs from January 11 to August 16, 2024.

ETFs and capitalizing on strong upward trends. This approach, however,

comes with heightened volatility, as evidenced by sharp fluctuations and occasional steep declines when trends reverse. In contrast, the Buy-and-Hold strategy follows a more moderate growth trajectory, benefiting from the long-term appreciation of the cryptocurrency market. While it endures significant short-term volatility, investors using this approach must be willing to tolerate market drawdowns, knowing that, historically, crypto assets have trended upward over extended periods, providing reassurance and confidence in this strategy. Meanwhile, the DCA strategy presents the most stable investment path. It gradually accumulates positions over time to mitigate the impact of market downturns. Unlike Buy-and-Hold, which exposes the investor to the entire market risk at all times, DCA smooths out volatility, resulting in a less erratic performance curve. Although this method sacrifices some potential upside during bull markets, it provides greater resilience during corrections, making it a safer and more secure choice for risk-averse investors.

It's also important to note that several transaction fees are associated with investing in cryptocurrency exchange-traded funds (ETFs), which may affect total returns. These expenses fall into two categories: explicit costs, which include direct fees and commissions, and implicit costs, which encompass market-related expenses such as slippage and spreads.

The expense ratio shows the ETF provider's recurring management fee. The complexity of managing cryptocurrency assets, custody solutions, and regulatory compliance sometimes results in expense ratios for crypto ETFs greater than those for standard ETFs, usually ranging from 0.25% to 1.5% annually. Crypto ETFs provide regulatory control and ease compared to direct cryptocurrency ownership, but they also have transaction costs that can reduce profits. Although they still suffer annual expenditure ratios, buy-and-hold investors are less impacted by bid-ask spreads and trading charges. On the other hand, because of spreads, slippage, and frequent transactions, active traders incur higher costs. Understanding these expenses is crucial for selecting the optimal ETF strategy and maximizing long-term returns.

8. Conclusion

In an era where cryptocurrencies have become one of the most significant and popular topics in the financial world, crypto ETFs have emerged as a focal point of one of the most exciting, innovative, and controversial developments in this field. The term ETF refers to exchange-traded funds that act as a bridge between the stock market and various assets. Cryptocurrencies

have led the world to a new concept of currencies and types of transactions, but owning and investing in this asset has often been accompanied by challenges. For this reason, the crypto ETFs, designed and offered by reputable companies and financial professionals, serve as a credible vehicle for investors to participate indirectly in cryptocurrencies by purchasing a unit of this fund.

This study employs machine learning classification algorithms to investigate the predictability of eight significant cryptocurrency exchange-traded funds across four distinct time scales: daily, 15-minute, 30-minute, and 60-minute returns. Numerical experiments using classification methods such as Logistic Regression, Linear Regression, Support Vector Machines, Artificial Neural Networks, K-means Clustering, and Random Forest algorithms demonstrate the predictability of upward or downward price movements.

By comparing the accuracy values across models, the Neural Networks and SVM models exhibit the highest accuracy values, often exceeding 0.80, with some values reaching 0.86. These models consistently perform well across different time frames and conditions. The Decision Tree and K-means Clustering models exhibit moderate accuracy, typically ranging from 0.55 to 0.78, which makes them less reliable for high-precision predictions. While struggling with non-linear data patterns, the Linear Regression and Logistic Regression models offer an advantage in model transparency and interpretability, which you can appreciate. Their accuracy levels generally range from 0.75 to 0.78.

Moreover, we compared the methods using the t-test and signed-rank test. Additionally, we discussed strategies in the crypto ETF market and concluded that the best approach depends on an investor's risk tolerance and objectives. If maximizing returns is the priority and one can withstand sharp drawdowns, the Momentum strategy offers the most significant growth potential. If an investor seeks long-term gains with moderate risk exposure, the Buy-and-Hold approach is a solid and reassuring choice. DCA provides the most balanced approach for those prioritizing stability and a methodical entry into the market, effectively mitigating the unpredictable nature of cryptocurrency markets. The findings of this study have important implications for market efficiency. First, if the crypto ETF market were efficient in the weak-form, its past price movements would not help us forecast future returns. However, our findings suggest that applying machine learning tools to historical prices is a game-changer, enabling the successful prediction of returns at the intraday level. This technological advancement is significant for algorithmic trading, as it successfully automates decisions on when

to buy and sell cryptocurrencies within a day. The success of such newly developed algorithms may also lead to more high-tech, dominant financial markets, marking a new era in the industry.

This inefficiency can also benefit institutional investors with access to sophisticated data analytics and trading tools, allowing them to exploit market anomalies more effectively. However, it's crucial to note that unique risks are present in the context of crypto ETFs, where the market is still evolving and regulatory clarity is developing. These risks, coupled with the potential for prolonged inefficiencies, may lead to increased volatility and higher risks for passive investors. We can significantly improve efficiency as the cryptocurrency market evolves and institutional adoption increases. The unique characteristics of digital assets, such as high volatility, fragmented liquidity, and changing regulations, indicate that some inefficiencies may persist longer than we would like. However, embracing these challenges can drive innovation and ultimately build a stronger and more resilient market.

9. Acknowledgments

While preparing this work, the authors used AI tools to check, correct, and organize text for more readability and precision.

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Appendix A.

The [Appendix A](#) includes t-test results and Wilcoxon's rank test for comparing all methods.

Table A.1: t test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

t-test	ARIMA vs Random Walk	ARIMA vs Linear Regression	ARIMA vs SVM	ARIMA vs Decision Tree	ARIMA vs Logistic Regression	ARIMA vs K-means	ARIMA vs Neural Networks
15 min 0.8	-4.65 (***)	-3.31 (***)	-3.53 (***)	-2.19 (**)	-3.48 (***)	-0.36	-2.62 (**)
15 min 0.9	-3.29 (**)	-2.71 (**)	-2.84 (**)	-1.08	-2.67 (**)	3.42 (***)	-2.78 (**)
60 min 0.8	-1.95 (*)	-1.94 (*)	-2.10 (*)	-0.97	-1.98 (*)	2.07 (*)	-2.15 (*)
60 min 0.9	0.79	-2.70 (**)	-3.52 (***)	-0.49	-3.59 (***)	2.61 (**)	-2.45 (**)
30 min 0.8	-6.92 (***)	-3.10 (**)	-3.57 (***)	-2.50 (**)	-3.43 (***)	-0.30	-3.33 (***)
30 min 0.9	1.32	-1.90 (*)	-1.86	-0.47	-1.96 (*)	3.08(**)	-1.83
Daily 0.8	-29.04 (***)	-2.68 (**)	-3.50 (***)	-1.91 (*)	-3.87 (***)	-2.24 (*)	-2.72 (**)

The values in the parentheses are t -statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively

Table A.2: t test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

t-test	Linear Regression vs Decision Tree	Linear Regression Logistic Regression	Linear Regression vs K-means	Linear Regression vs Neural Networks	SVM vs Decision Tree	SVM vs Logistic Regression	SVM vs K-means
15 min 0.8	1.64	-0.05	3.48 (***)	0.25	1.78 (*)	0.02	3.75 (***)
15 min 0.9	1.60	0.01	3.85 (***)	-0.03	1.68	0.06	4.00 (***)
60 min 0.8	1.09	0.01	3.05 (**)	-0.05	1.18	0.04	3.28 (**)
60 min 0.9	1.66	-0.28	3.63 (***)	0.02	2.06 (*)	-0.03	4.41 (***)
30 min 0.8	1.03	-0.11	2.55 (**)	-0.08	1.26	0.04	2.89 (**)
30 min 0.9	1.43	-0.02	2.73 (**)	0.03	1.39	-0.04	2.68 (**)
Daily 0.8	1.18	-0.35	0.70	-0.05	1.63	-0.06	1.07

The values in the parentheses are t-statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively

Table A.3: t test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

t-test	SVM vs Neural		Decision Tree vs		Decision Tree vs		Logistic Regression		Logistic Regression		Logistic Regression		K-means vs		K-means vs	
	Network		vs K-means		Neural Networks		vs K-means		vs K-means		vs Neural Networks		Neural Networks		Neural Networks	
15 min 0.8	0.32		-1.75	2.44 (**)	-1.17		3.69 (***)		0.30		-2.67 (**)					
15 min 0.9	0.01		-1.58	2.70 (**)	-1.65		3.81 (***)		-0.05		-3.92 (***)					
60 min 0.8	-0.02		-1.10	2.47 (**)	-1.21		3.12 (**)		-0.06		-3.35 (***)					
60 min 0.9	0.26		-2.09 (*)	1.72	-1.56		4.47 (***)		0.29		-3.36 (***)					
30 min 0.8	0.07		-1.20	1.81 (*)	-1.15		2.78 (**)		0.03		-2.71 (**)					
30 min 0.9	0.01		-1.47	1.90 (*)	-1.38		2.80 (**)		0.05		-2.65 (**)					
Daily 0.8	0.22		-1.78 (*)	-0.50	-1.23		1.18		0.29		-0.75					

The values in the parentheses are t-statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively

Table A.4: Wilcoxon signed rank test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

signed rank	ARIMA vs Random Walk	ARIMA vs Linear Regression	ARIMA vs SVM	ARIMA vs Decision Tree	ARIMA vs Logistic Regression	ARIMA vs K-means	ARIMA vs Neural Networks
15 min 0.8	(1.0), p = 0.0156 **	(4.0), p = 0.0547 *	(4.0), p = 0.0547 *	(7.0), p = 0.1484	(4.0), p = 0.0547 *	(15.0), p = 0.7422	(7.0), p = 0.1484
15 min 0.9	(0.0), p=0.0078 ***	(4.0), p = 0.0547 *	(2.0), p = 0.0234 **	(12.0), p = 0.4609	(4.0), p = 0.0547 *	(0.0), p = 0.0078 ***	(3.0), p = 0.0391 **
60 min 0.8	(15.0), p=0.7422	(7.0), p = 0.1484	(4.0), p=0.0547 *	(8.0), p = 0.1953	(6.0), p = 0.1094	(6.0), p = 0.1094	(3.0), p = 0.0391 **
60 min 0.9	(9.0), p = 0.2500	(4.0), p = 0.0547 *	(0.0), p = 0.0078 ***	(16.0), p = 0.8438	(0.0), p = 0.0078 ***	(5.0), p = 0.0781 *	(4.0), p=0.0547 *
30 min 0.8	(1.0), p = 0.0156 **	(4.0), p = 0.0547 *	(2.0), p = 0.0234 **	(4.0), p = 0.0547 *	(3.0), p = 0.0391 **	(17.0), p=0.9453	(2.0), p = 0.0234 **
30 min 0.9	(3.0), p = 0.0391 **	(7.0), p = 0.1484	(7.0), p = 0.1484	(16.0), p = 0.8438	(8.0), p = 0.1953	(1.0), p = 0.0156 **	(7.0), p = 0.1484
Daily 0.8	(0.0), p = 0.0078 ***	(3.0), p = 0.0391 **	(2.0), p = 0.0234 **	(5.0), p = 0.0781 *	(0.0), p = 0.0078 ***	(7.0), p = 0.1484	(5.0), p = 0.0781 *

The values in the parentheses are t-statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively

Table A.5: Wilcoxon signed rank test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

signed rank	Linear Regression vs Decision Tree	Linear Regression vs Logistic Regression	Linear Regression vs K-means	Linear Regression Neural Networks	SVM vs Decision Tree	SVM vs Logistic Regression	SVM vs K-means
15min 0.8	(1.0), p=0.0156 **	(5.0), p=1.0000	(1.0), p=0.0156 **	(7.0), p=0.8927	(0.0), p=0.0078 ***	(0.0), p=0.1088	(0.0), p=0.0078 ***
15min 0.9	(0.0), p=0.0078 ***	(1.0), p=0.2850	(0.0), p=0.0078 ***	(3.0), p=0.4652	(0.0), p=0.0000	(0.0), p=0.0000	(0.0), p=0.0078 ***
60min 0.8	(2.0), p=0.0234 **	(3.0), p=1.0000	(3.0), p=0.391 **	(2.0), p=0.5930	(0.0), p=0.0000	(1.0), p=0.2850	(0.0), p=0.0180 ***
60min 0.9	(1.0), p=0.0156 **	(0.0), p=0.0679 *	(1.5), p=0.0234 **	(4.0), p=0.7150	(0.0), p=0.0000	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***
30min 0.8	(3.0), p=0.0391 **	(2.0), p=0.2733	(1.1), p=0.2850	(3.0), p=0.4652	(0.0), p=0.0000	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***
30min 0.9	(3.0), p=0.0391 **	(1.0), p=0.6547	(1.0), p=0.0156 **	(0.0), p=0.1797	(4.0), p=1.0000	(1.0), p=0.0156 **	(0.0), p=0.0178 **
Daily 0.8	(5.0), p=0.0781 *	(1.0), p=0.2850	(8.5), p=0.2500	(7.0), p=0.8927	(0.0), p=0.0000	(0.0), p=0.0178 **	(0.0), p=0.0178 **

The values in the parentheses are t-statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively

Table A.6: Wilcoxon signed rank test results for comparison of classification algorithms and time series methods over different time scales from January 11 to August 16, 2024.

signed rank	SVM vs Neural		Decision Tree vs		Decision Tree vs		Logistic Regression vs		Logistic Regression vs		K-means vs	
	Network		Logistic Regression	vs K-means	Neural Networks	vs K-means	Neural Networks	vs K-means	Neural Networks	vs K-means	Neural Networks	vs K-means
15min 0.8	(4.0), p=0.7150	(1.0), p=0.0156 **	(3.0), p=0.0391 **	(4.0), p=0.0547 *	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(3.0), p=0.0391 **	(0.0), p=0.0078 ***
15min 0.9	(7.0), p=0.8927	(4.0), p=0.0547 *	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***
60min 0.8	(0.0), p=0.1797	(1.0), p=0.2850	(3.0), p=0.0630 *	(1.0), p=0.0156 **	(0.0), p=0.0078 ***	(1.0), p=0.0156 **	(3.0), p=0.0391 **	(1.0), p=0.0156 **	(3.0), p=0.0391 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **
60min 0.9	(1.0), p=0.2850	(0.0), p=0.1094	(6.0), p=0.0234 **	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(2.0), p=0.0234 **	(2.0), p=0.0234 **
30min 0.8	(0.0), p=0.1088	(0.0), p=0.0180 **	(3.0), p=0.391 **	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***	(0.0), p=0.0078 ***
30min 0.9	(0.0), p=0.3173	(3.0), p=0.091 *	(1.0), p=0.6547	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **	(1.0), p=0.0156 **
Daily 0.8	(4.0), p=0.7150	(1.0), p=0.1484	(6.0), p=0.3428	(4.0), p=0.7150	(4.0), p=0.7150	(4.0), p=0.7150	(4.0), p=0.7150	(4.0), p=0.7150	(2.0), p=0.0234 **	(2.0), p=0.0234 **	(6.0), p=0.1763	(6.0), p=0.1763

The values in the parentheses are t-statistics. ***, ** and * denote significance at the 1%, 5% and 10% level respectively